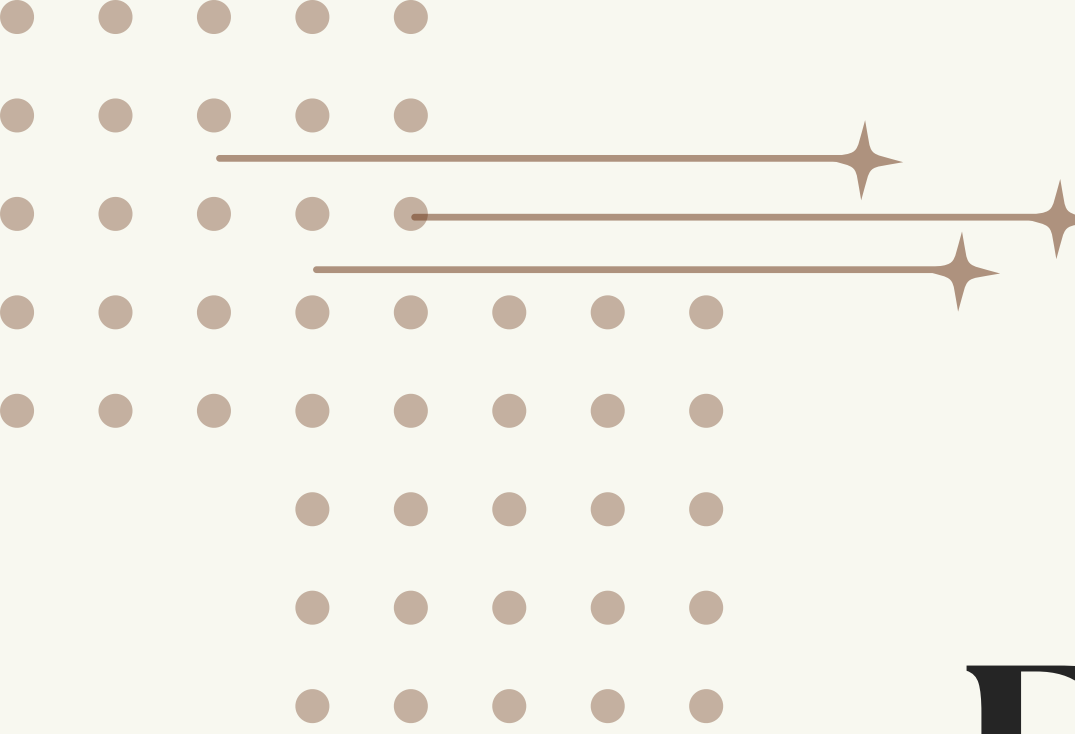
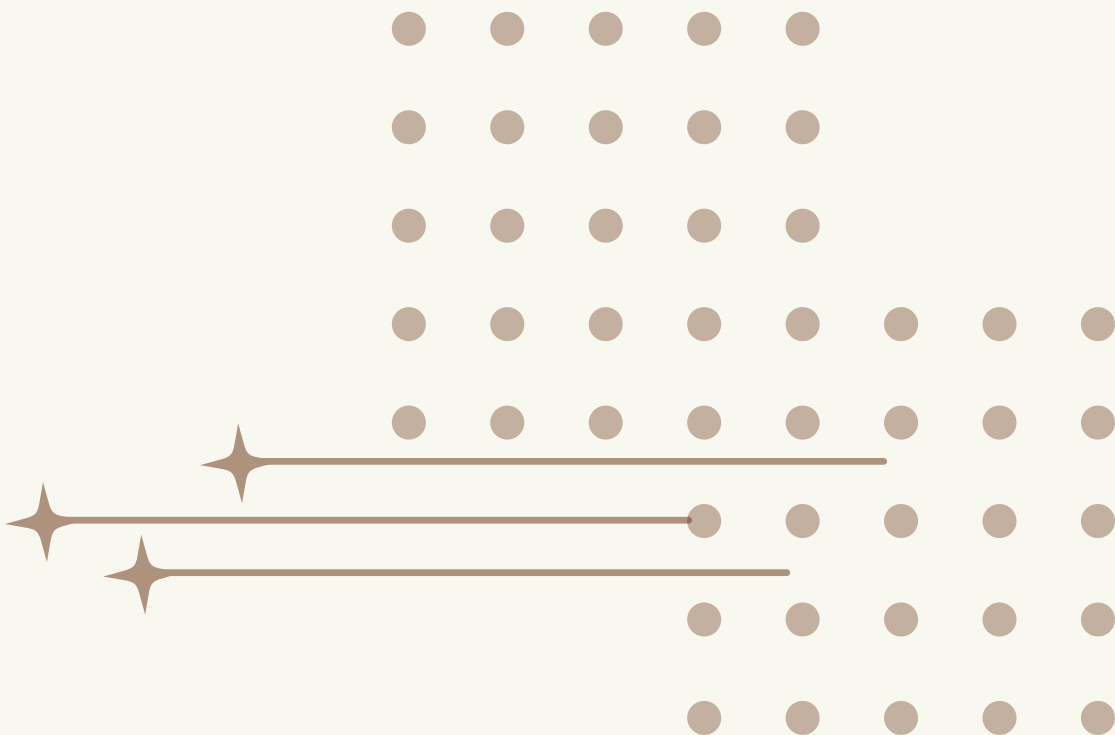


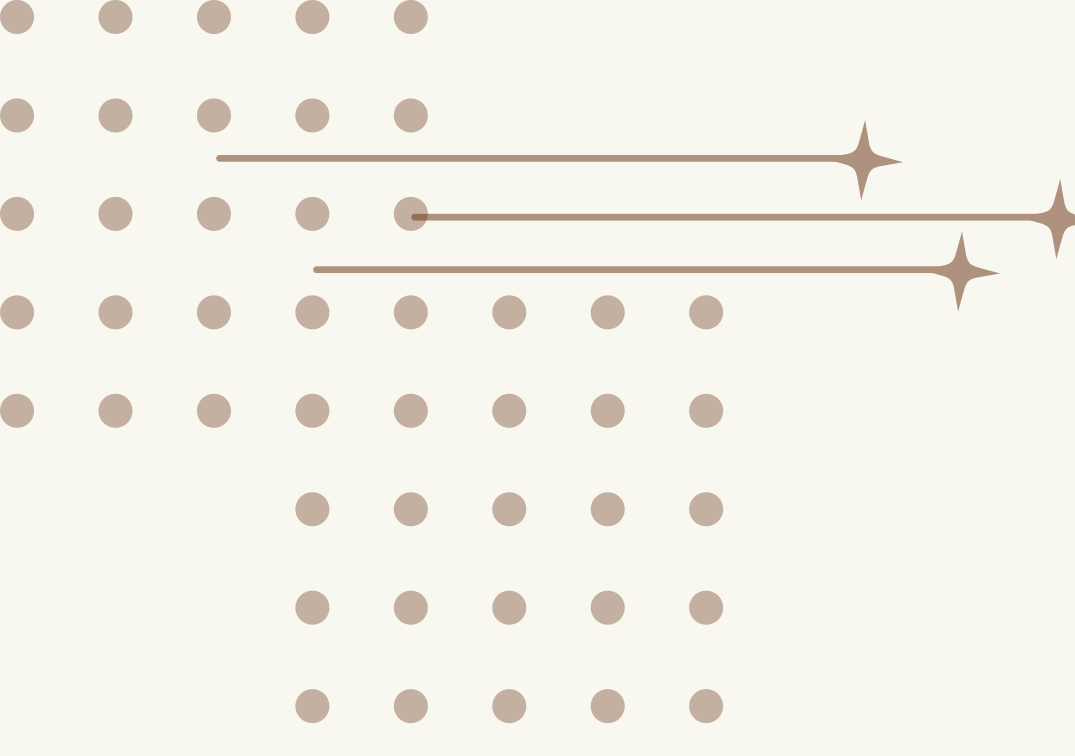


Thomas More

BACHELOR PORTFOLIO

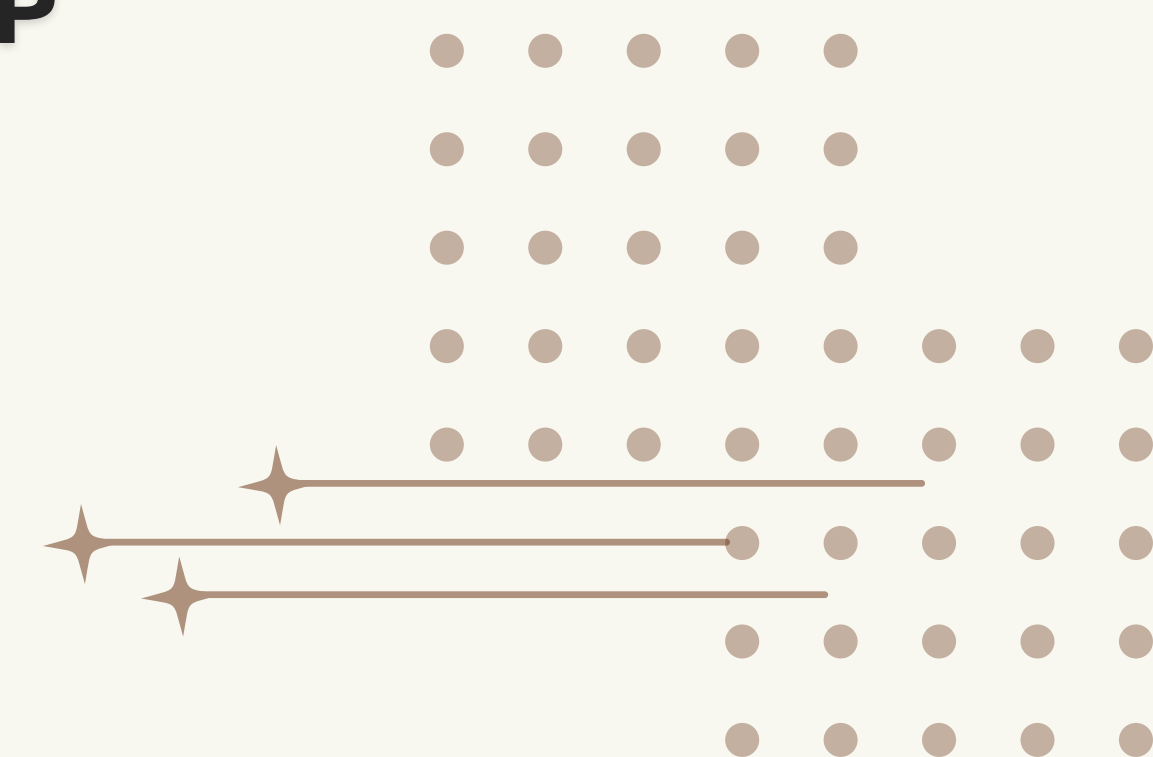
Julia Drygalska





OVERVIEW



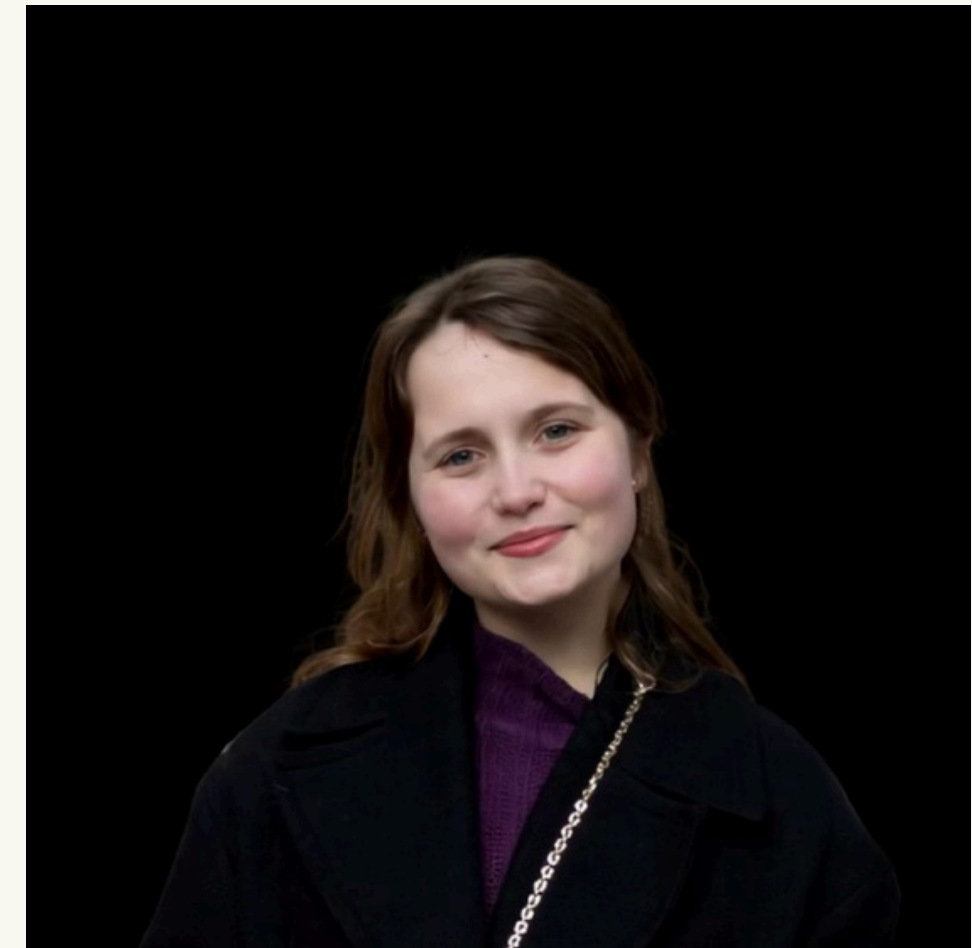
- **INTRODUCTION**
 - **INTERNSHIP**
 - **CERIQ - LLM APP**
 - **CHEF TUTOR - LLM APP**
 - **CONCLUSION**
- 

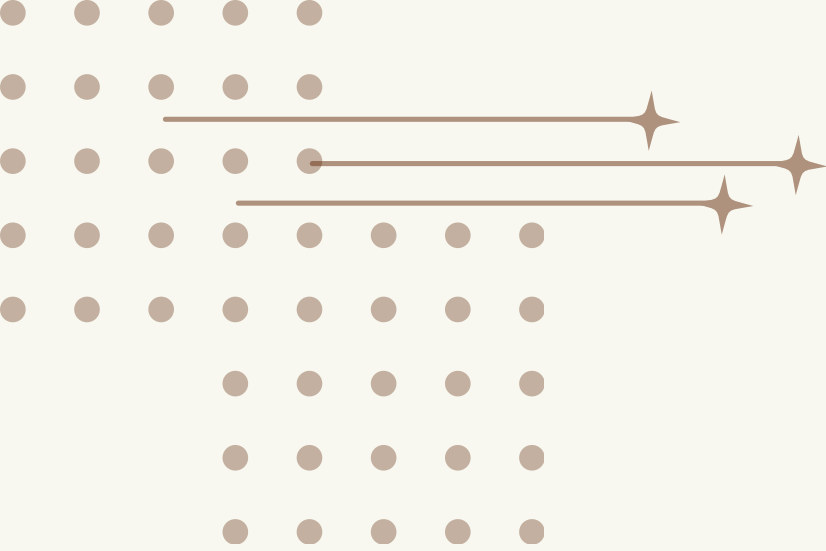
INTRODUCTION

Julia Drygalska

Applied Computer Science student
Thomas More University of Applied Sciences

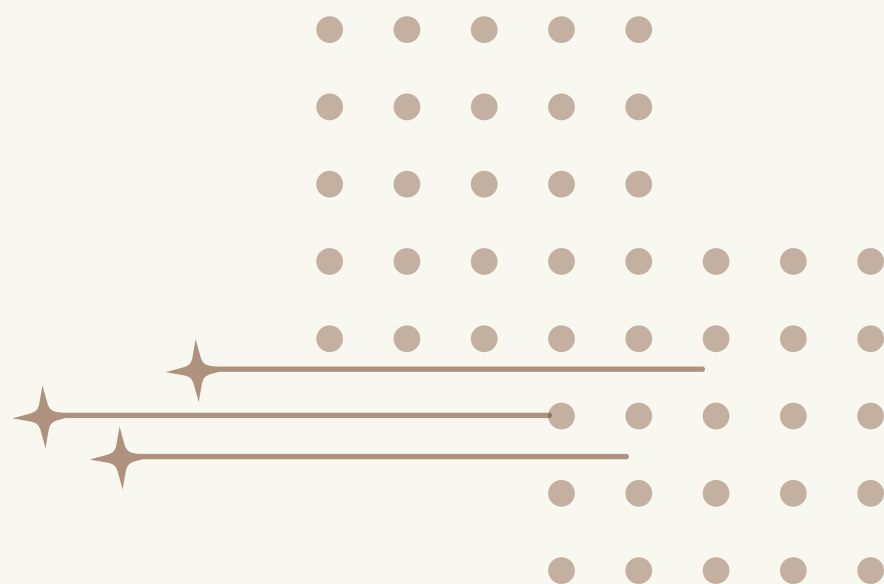
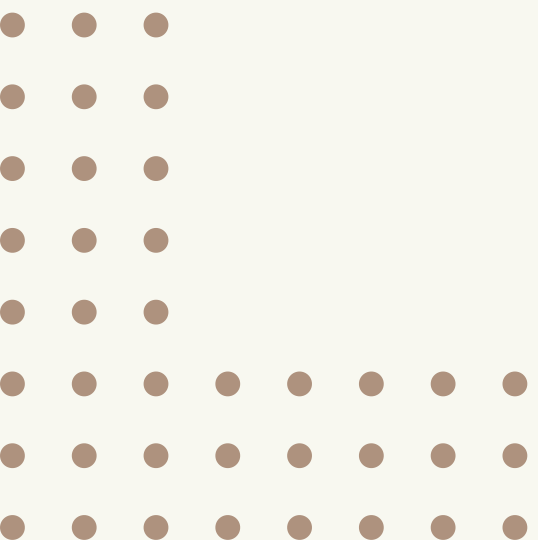
I am interested in AI, computer vision and
practical software solutions.

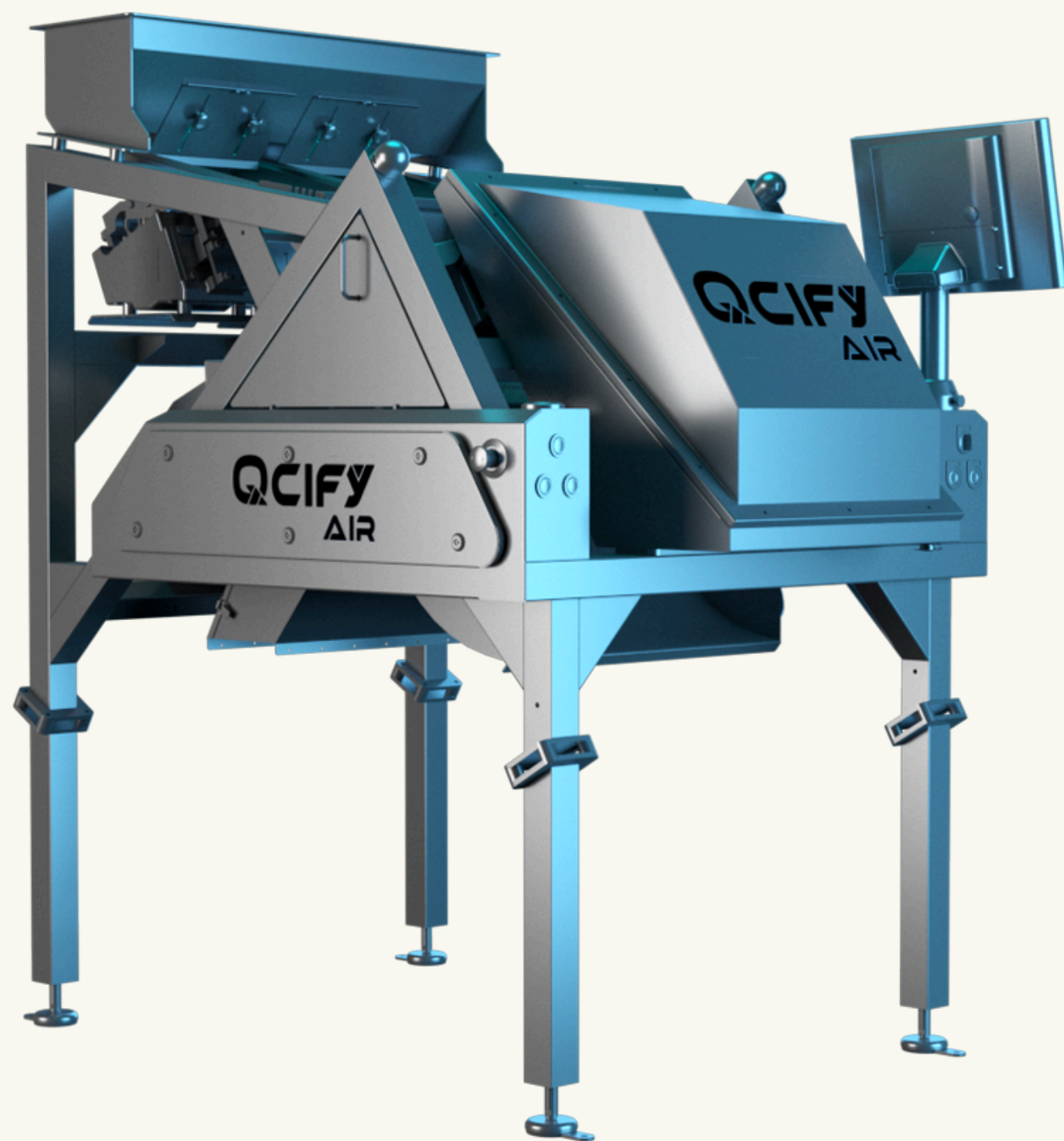




INTERNSHIP

QCIFY





RESEARCH

Goal: Improve almond declustering

Focus areas:

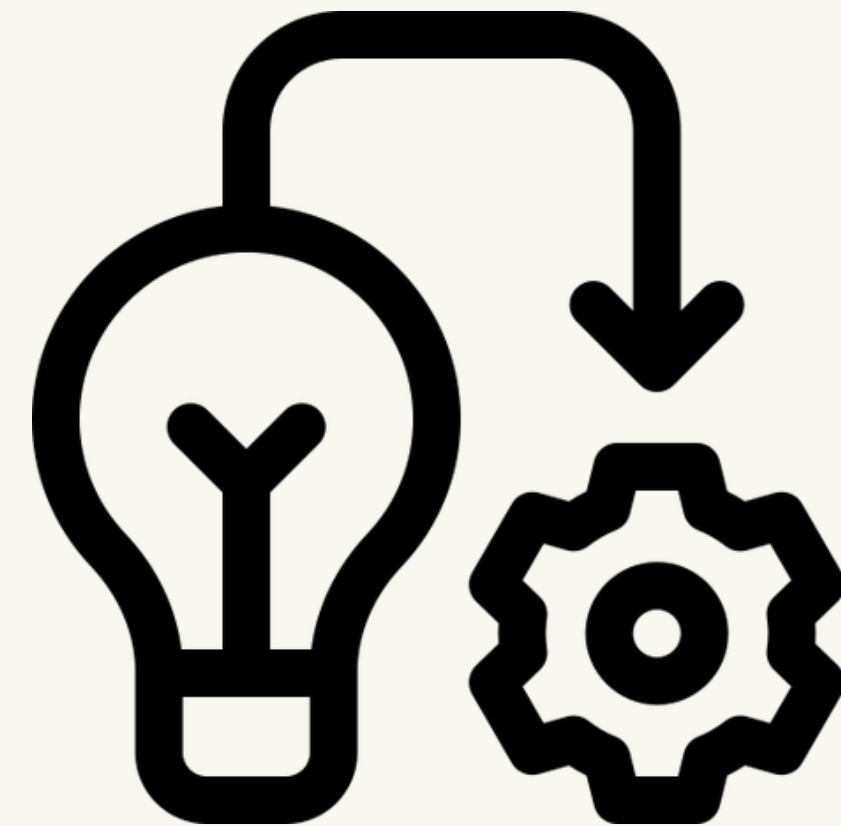
- Segmentation models
- Boundary detection
- Distance maps
- Model performance
- Runtime constraints



IMPLEMENTATION

From research to testing

- Prepared training data
- Created extra target maps
- Trained segmentation models
- Compared results



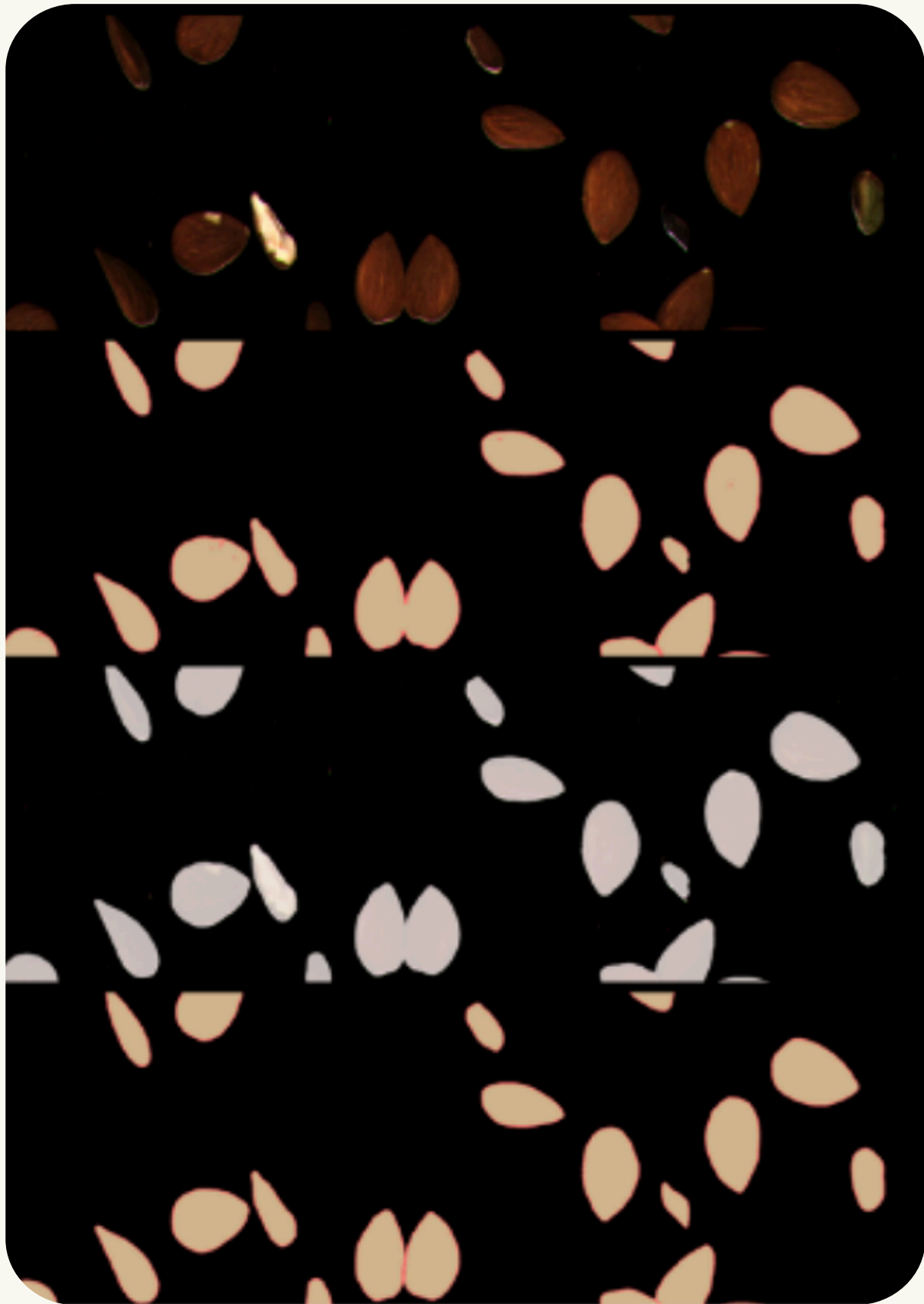
RESULTS & EVALUATION

- Are almonds separated correctly?
- Are objects counted accurately?
- Is the model fast enough?
- Do the results look reliable visually?

Key focus:

Accuracy + real-time performance





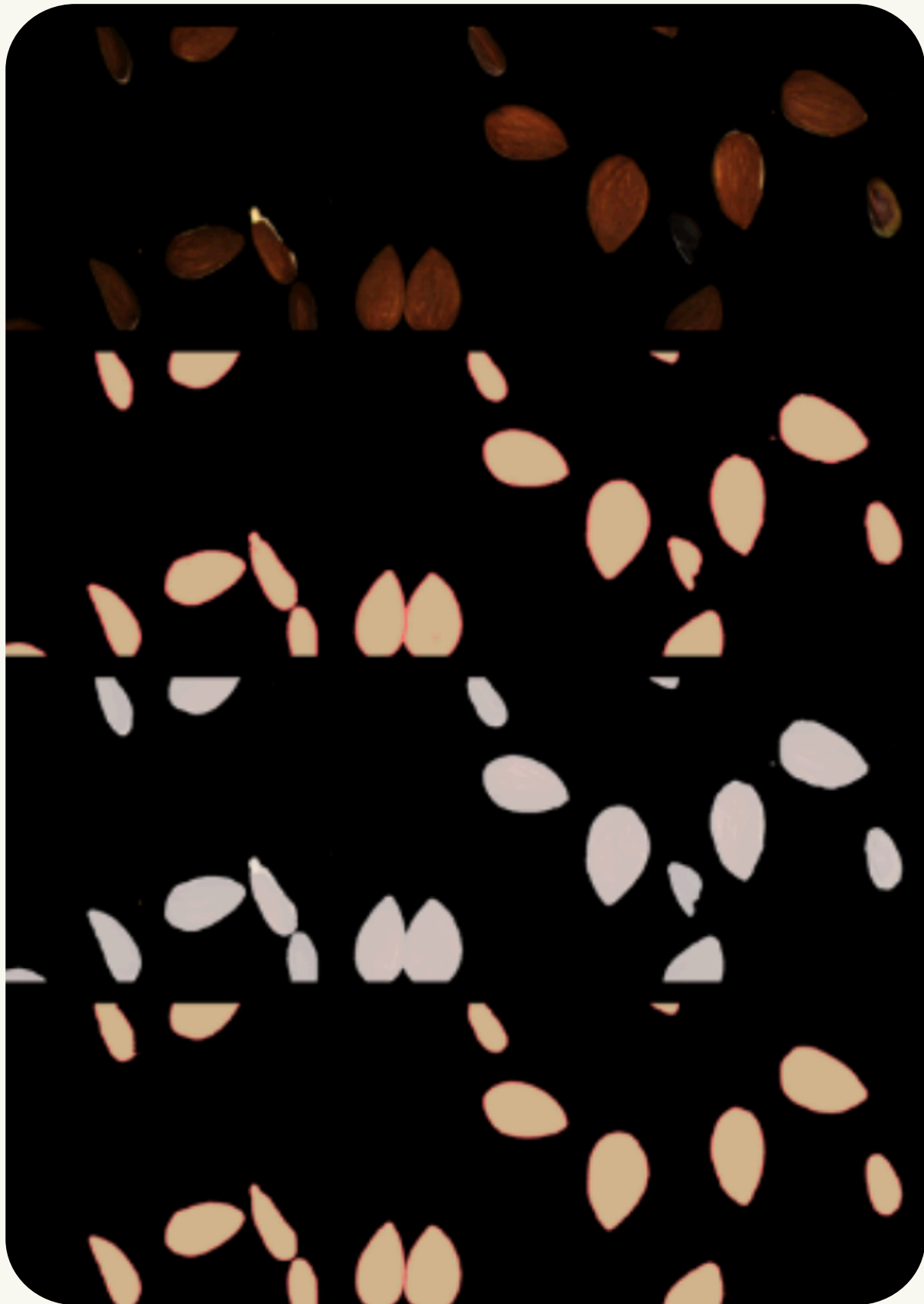
Test-04, 32 px

IoU: 0.9614

Count MAE: 0.2082

Merge rate: 0.1026

Split rate: 0.0670



Test-11, 64 px

IoU: 0.9711

Count MAE: 0.2884

Merge rate: 0.1199

Split rate: 0.1213



Test-18, 64 px

IoU: 0.9734

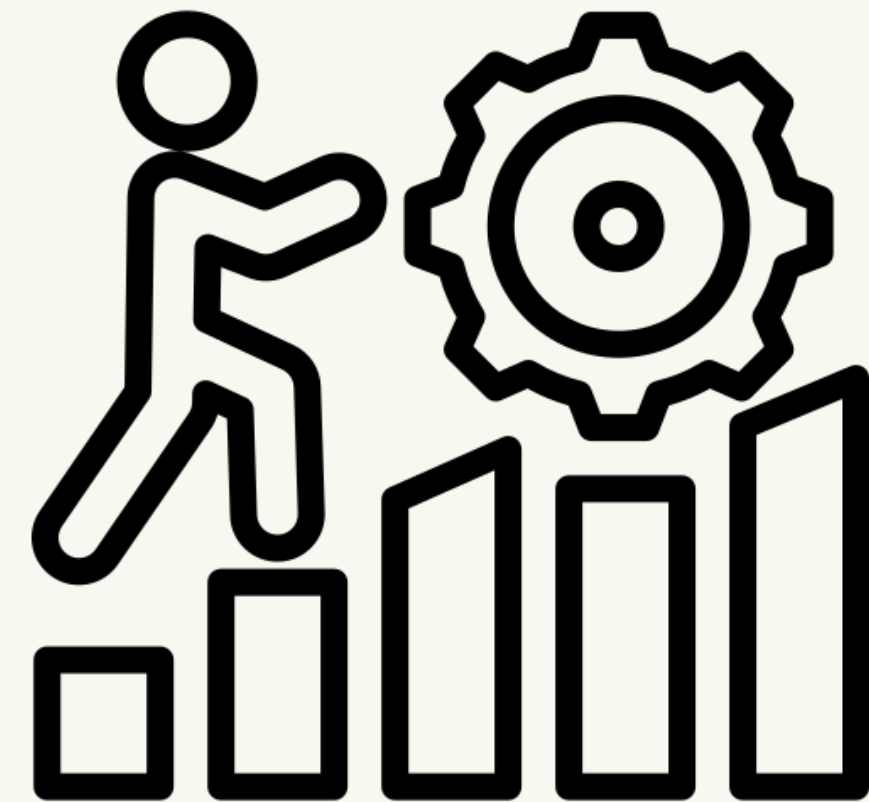
Count MAE: 0.2548

Merge rate: 0.0949

Split rate: 0.0892

CHALLENGES

- Merged objects
- Over-segmentation
- Balance between accuracy and runtime
- Choosing the right evaluation metrics
- Comparing visual results with numerical scores



CERIQ

Inbox

Q Search...

AL alex@nebula-design.com 1/28/2026, 09:32 PM

Manufacturing enquiry - Custom shape

✓ Ready

J. j.smith@hyperloop-tech.c... 1/28/2026, 06:08 PM

Question regarding technical report

✓ Ready

SA sarah.k@siemens-healthine... 1/28/2026, 04:38 PM

Re: Order #9921 status update needed ASAP

✓ Ready

+1

DA david.chen@luminx-lightin... 1/28/2026, 03:38 PM

Re: Thermal management for High Power UV-C ...

✓ Ready

+4

HI Hamid Iqbal 1/28/2026, 01:49 PM

Request for Definitions and Guidance on Absolute ...

Generating

M. m.schmidt@continental.com 1/28/2026, 01:38 PM

Re: RFQ: Pressure Sensor Substrate (Ceramic)

+2

Q Search...

AL alex@nebula-design.com 1/28/2026, 09:32 PM

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M. m.schmidt@continental.com 1/28/2026, 01:38 PM

Re: RFQ: Pressure Sensor Substrate (Ceramic)

+2

JA james.neller@arrow.com 1/28/2026, 12:03 PM

Distribution partnership for CERcuts in North Ame...

+2

DA dr.huber@photonics-lab.de 1/28/2026, 11:28 AM

Question regarding laser drilling via

+2

Manufacturing enquiry - Custom shape

AL alex@nebula-design.com

Wed, Jan 28, 2026, 09:32 PM

Hi,

We are looking for a supplier for a new wearable project.
Can you make a ceramic PCB shaped exactly like this?

See attached image for the reference design.

Kind regards,
Alex
Nebula Design Studio

ATTACHMENTS

Image_28-01-2026_at_10.19.webp (488 KB)

AI Generated Response Version 1 of 1

Subject: Ceramic PCB Customization Feasibility – Based on Your Reference Design

Hi Alex,

Thanks for sharing the reference image. Based on the visual analysis, we can confirm that the board features a custom shape with gold-plated areas, through-hole pins, and irregular edges – all of which are manufacturable with ceramic PCB technology [1]. The gold plating is consistent with high-reliability applications, which aligns with our capabilities for surface finishes like Au [1].

Findings:

- The board's shape, including curved/irregular edges, is feasible for ceramic PCB fabrication [1].
- Gold plating on conductive areas (e.g., central rectangle, edge pins) is supported [1].
- Through-hole pins and mounting holes are standard features we can accommodate [1].
- No dimension annotations or tolerances are visible in the image – these are required for manufacturing [4].
- The design appears to be single- or double-sided, which is within our standard process range [1].

General engineering note:

When designing custom-shaped ceramic PCBs, it's critical to avoid overlapping or protruding lines in the Gerber files – these can cause conversion errors or misinterpretation during fabrication [4]. Also, ensure all metallized areas are clearly defined (e.g., via shading or layer labeling) to avoid ambiguity.

To proceed:

- Could you please share the Gerber files or a design drawing (with dimensions and tolerances) for the exact shape and layout?
- What substrate material are you targeting? (e.g., AlN, Al2O3, Si3N4) – this affects thermal performance and manufacturability.
- Are there any specific thickness or layer count requirements?
- Do you need any special finishes (e.g., AuSn solder pre-deposits, Pt barriers) or additional features (e.g., laser cutouts, castellated edges)?

We're happy to review your design and provide a quote once we have these details.

Kind regards,
CERcuts Technical Team

Sources: attachments_context, Invoice-2025-03.pdf, ECSS-Q-01-75-4103(Apr2022).pdf, Substrate-Design-Guide-USA-190409-2.pdf

10:06 PM - 28 Jan 2026

1

Edit Regenerate

+ New Chat

Hi there 1/28/2026

Hey, so could you tell ... 1/28/2026

Could you tell me why... 1/28/2026

design_schematic.png

Hi there

Hi David,

Thank you for your email and the attached schematic. I'm sorry for any confusion, but it seems there might have been a mix-up. The schematic you provided appears to be for an audio amplifier circuit, not a UV-C LED module as initially discussed.

To ensure we provide the most accurate assistance, could you please:

1. Confirm if the attached schematic is indeed for the UV-C LED module project we discussed.
2. If not, could you please share the correct schematic and board outline for the UV-C LED module?

We look forward to your clarification so we can proceed with verifying the thermal pad size for your Alumina substrate.

Best regards,
[Your Name]
CERcuts Technical Support Team

What source documents were used for the email you had before? and what inside them, give me a list

The sources referenced in the previous email response include the following documents and their relevant content:

1. ****attachments_context**** ([20000])
 - This source likely contains the schematic and board outline files attached by David Chen. The content includes the design_schematic.png file, which was reviewed and found to be an audio amplifier circuit instead of a UV-C LED module.
2. ****thread_context**** ([20001])
 - This source contains the email thread between David Chen and the CERcuts Technical Team. It includes the initial email from David Chen discussing the thermal

Type a message...

CHEF TUTOR

 **Upload recipe PDFs**

Add new recipe PDFs (PL / ES / EN)

Drag and drop files here

Limit 200MB per file • PDF

Browse files

☐ Rebuild vector DB from scratch (delete old entries) ?

Ingest uploaded PDFs

Deploy

 Can you give me Peruvian–Polish fusion dumpling with sauce

 You're looking for a new recipe. Let's combine the flavors of Poland and Peru to create a unique dish.

How about "Pierogi de Pollo con Aji Amarillo y Salsa de Mostaza" (Polish-Peruvian Chicken Dumplings with Ají Amarillo Sauce and Mustard Sauce)?

Here's a recipe:

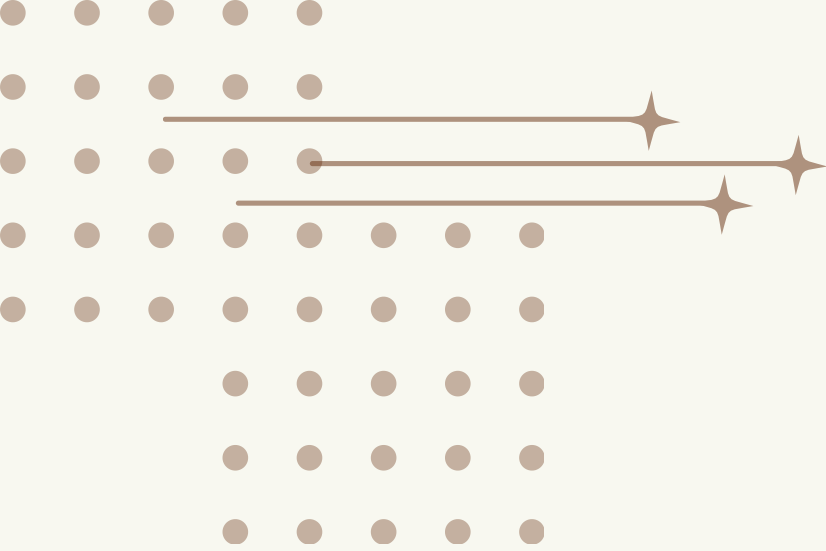
Ingredients:

- 500g all-purpose flour
- 300g warm water
- 1/2 teaspoon salt
- Filling:
 - 250g cooked chicken, diced
 - 100g grated Polish cheese (e.g., Oscypek)
 - 50g sautéed onions and garlic
 - 20g chopped fresh cilantro

Ask for a recipe, fusion idea, macros, 'convert 200 g to oz', 'quiz', or 'history'...

CONCLUSION





THANK YOU

